

Non-negative Tensor Factorization based Bi-Clustering on Anchor Graphs

Quanxue Gao, Rui Wang, Jing Li, Xiaoke Ma, Cheng Deng, Yu Duan

Abstract—Non-negative matrix factorization (NMF) is a powerful technique for clustering analysis. Applying it to multi-view clustering, the existing methods typically apply NMF to obtain soft label matrices for each view independently and then fuse the obtained label matrices of views to generate view-consistent label matrix. A major drawback is that they mainly focus on decision-level fusion and can't effectively exploit the implicit relationships between views at the data level, which are critical for multi-view clustering. Moreover, they rely on the relationship between NMF and K-means to derive labels, which has weak interpretability. To address these problems, we propose a novel multi-view clustering model based on non-negative tensor factorization (NTF), which employs multi-level fusion (both data-level and decision-level) to achieve view-consistent labels for multi-view data. Specifically, we present a non-negative tensor factorization and utilize it to decompose tensorized anchor graph into the product of an anchor indicator tensor and a data indicator tensor from a probabilistically interpretable perspective, thereby enhancing the interpretability of our proposed model. Extensive experimental results demonstrate the effectiveness of our method.

Index Terms—Non-negative Tensor Factorization, Multi-view Clustering, Anchor Graph, Bi-clustering

1 INTRODUCTION

CLUSTERING, a fundamental technique in unsupervised learning, is widely employed to categorize samples into distinct groups or clusters. In the burgeoning field of big data, the acquisition of vast volumes of unlabeled data has become increasingly straightforward. Clustering plays a pivotal role in uncovering the intrinsic structures and patterns in such data, thereby facilitating the assignment of labels to unlabeled data, which is crucial for enhancing the utility of these data in downstream tasks [1]–[4]. Consequently, clustering, especially multi-view clustering (MVC) has emerged as a prominent area of research in machine learning [5]–[7].

Among the various MVC technologies [7]–[10], graph-based multi-view clustering stands out as a benchmark approach due to its ability to handle data points with arbitrary shapes. However, graph-based methods involve $n \times n$ graph construction and eigen-decomposition of the Laplacian matrix, with computational complexities of $\mathcal{O}(Vn^2)$ and $\mathcal{O}(n^3)$, respectively. These requirements limit their applicability to large-scale multi-view data. To address this, a myriad of anchor graph-based clustering methods have been proposed [11]–[13]. Anchor graphs represent multi-view data by modeling the relationship between n samples and m anchors, effectively reducing computational and storage complexities due to $m \ll n$. Generally, to avoid

post-processing, some multi-view clustering methods based on anchor graphs aim to learn a bipartite graph with K -connected components and derive clustering labels directly from these connected components [14]–[16]. These methods directly yield K clusters based on K -connected components, eliminating the need for additional post-processing. However, such methods often have stringent parameter requirements and may struggle to achieve K -connected components in certain cases. Another notable drawback is their lack of interpretability.

To address this, an effective and efficient strategy is to leverage non-negative matrix factorization (NMF) [17] on the anchor graph to directly obtain the cluster indicator matrix [18]–[20]. Most existing NMF-based multi-view clustering methods decompose anchor graph into cluster centroids and the corresponding soft indicator matrices, i.e. representation by using the relationship between NMF and K -means. This helps enhance the interpretability and achieve good and stable clustering performance. However, all of them face several challenges: First, they perform NMF on each view to learn the corresponding indicator matrix and learn the view-shared label matrix by decision-level fusion. This can't effectively exploit the implicit relationships between views at the data level, which is important for clustering. Second, they overlook the intrinsic geometric structure embedded in anchor graph. These challenges motivate us to explore whether a reasonable strategy can be used to alleviate these problems.

Multiple works [21], [22] have demonstrated that information embedded in across views helps to improve the representation ability of samples. Similarly, information embedded in across views should be leveraged to learn better indicator matrix from anchor graphs. Specifically, using information across views (data-level fusion) and ensuring the view-consistent of indicator matrices of views (decision-level) will facilitate the learning of indicator matrix. Furthermore,

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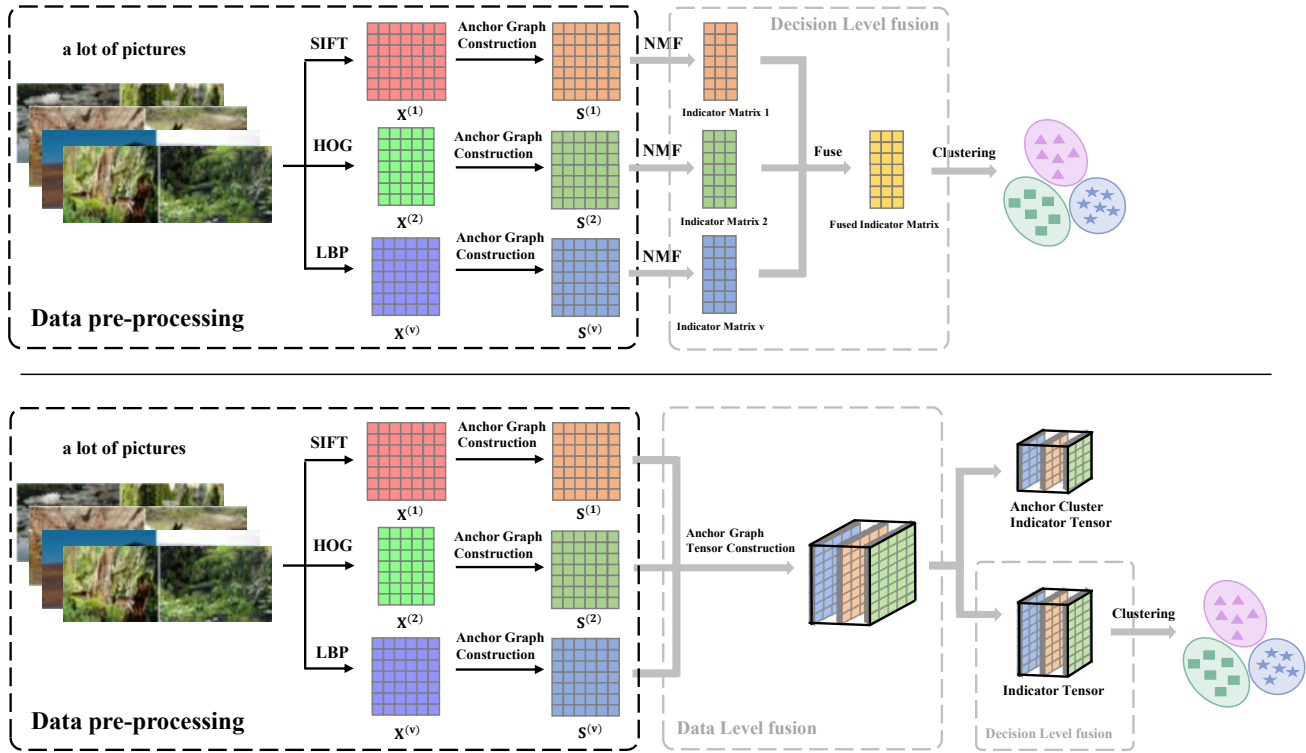


Fig. 1. The framework of conventional NMF-based multi-view clustering methods (top) and our proposed tensor-based framework (bottom). Conventional NMF-based methods only perform decision-level fusion across multiple views, whereas our method constructs a third-order tensor by stacking multiple anchor graph matrices along the third dimension and directly conducts tensor decomposition on it. This design enables simultaneous data-level and decision-level fusion, thus capturing cross-view correlations more effectively.

the probability characteristic of anchor graph should be taken into account in the learning procedure of indicator matrix. However, this characteristic is not considered in the existing anchor-based multi-view clustering methods, and our strategy is proposed based on it.

In this work, we propose a novel multi-view clustering model based on non-negative tensor factorization, which employs multi-level fusion (both data-level and decision-level) to achieve the view-consistent labels. As illustrated in Fig. 1, most existing NMF-based multi-view clustering methods perform clustering through decision-level fusion by independently factorizing multiple views and then integrating the learned representations, while neglecting the latent correlations among different views at the data level. In contrast, our model directly stacks multiple anchor graph matrices along the third dimension to construct a third-order tensor, and then directly performs tensor decomposition on the tensor, enabling unified data-level modeling across views. Specifically, we present a non-negative tensor factorization (NTF) and utilize NTF to decompose tensorized anchor graph into the product of an anchor indicator tensor and a data indicator tensor from a probabilistical perspective, thereby enhancing the interpretability of our proposed model and similarly achieving data-level fusion. To learn the view-consistent labels for different views, we employ the tensor Schatten p -norm to enforce the low-rank constraint on both the anchor indicator tensor and data indicator tensor, effectively capturing complementary information between indicator matrices of views and achieving decision-level fusion. The primary contributions of our paper are summarized as follows:

- **Model Construction.** According to the probability characteristic of an anchor graph, we decompose tensorized anchor graph into the product of an anchor indicator tensor and a data indicator tensor by non-negative tensor factorization (NTF), and then build orthogonal constraint NTF clustering model which achieves both data-level and decision-level fusion, while existing models are not.
- **Bi-clustering.** Our proposed method incorporates tensor Schatten p -norm for both the sample and anchor indicator tensors, whereas the existing methods solely imposes on sample indicator tensor. This dual constraint approach enables the integration of information from diverse perspectives, effectively achieving bi-clustering on anchor graphs and exploiting complementary information across views.
- **Probabilistic Interpretability.** Anchor indicator tensor $\mathcal{G}$ is constrained such that the sum of the column elements of each frontal slice satisfies $\mathcal{G}^{(v)} \cdot \mathbf{1} = \mathbf{1}$. This means each element in a row of any frontal slice of $\mathcal{G}$ can be interpreted as the probability that the corresponding anchor belongs to a specific cluster. The position of the largest value in a row indicates the cluster to which the anchor most likely belongs. This constraint enhances the interpretability of our proposed model from a probabilistic perspective.

2 RELATED WORKS

Using anchor graphs for multi-view clustering is a common approach. Multi-view clustering (MVC) based on anchor graphs is proposed to address the limitations of traditional

graph clustering methods, which struggle with large-scale multi-view data. Compared with $n \times n$ similarity graph, $n \times m$ ($m \ll n$) anchor graph significantly reduces computational and storage complexities [23].

One partial anchor graph-based strategy involves performing clustering techniques, such as spectral clustering and subspace clustering, on the constructed anchor graph to obtain clustering results [24]–[27]. However, this post-processing step is time-consuming, and the clustering performance is sub-optimal. Consequently, many MVC methods based on anchor graphs without post-processing have gradually been proposed.

To avoid post-processing, it is more common to construct bipartite graphs using anchor graphs, then learn bipartite graphs with K -connected components, and finally derive the clustering labels via the connected components [16], [28]. For example, MVSC [28] constructs sub-bipartite graphs in each view using raw data points and anchors and employs local manifold fusion to integrate the information from individual sub-graphs. Clustering labels of anchors are also utilized to handle out-of-sample problems. To reduce the number of parameters, SFMC [29] introduces a parameter-free multi-view clustering graph fusion model for obtaining compatible union graphs through self-supervised weighting. To better exploit complementary information across views, MSC-BG [14] employs the Schatten p -norm to capture complementary information between views and derives clusters through K -connected components. Similarly, TBGL [15] applies the Schatten p -norm to explore inter-view similarities while incorporating $\ell_{1,2}$ -norm minimization regularization and connectivity constraints to explore intra-view similarities. Although the aforementioned methods circumvent the need for post-processing with connected components, they often lack interpretability and may fail to identify K -connected components in certain cases.

Another commonly used technique for multi-view clustering is NMF or semi-NMF, which enhances the interpretability of models [17], [30], [31]. In the multi-view context, a straightforward strategy involves aligning the coefficients matrices from different views to ensure similarity or identity, forming the foundation of NMF-based MVC methods [24]. To integrate multi-view data within the NMF paradigm, Akata et al. [32] enforced identical clustering index matrices across different views via NMF decomposition. Recognizing the incomparable scales and difference among views, Liu et al. [33] performed NMF independently on each view, followed by aligning the resultant clustering index matrices to establish cross-view comparability.

To well handle large-scale multi-view data, inspired by SNMF/NMF-based MVC framework, many efficient clustering methods have been presented. For example, Kang et al. [26] learned anchor graph for each view according to anchors of the corresponding view, avoiding the construction of full graphs. However, the anchor points between different views are independent of each other and cannot effectively explore complementary information between views. To overcome this drawback, wang et al. [25] simultaneously learn both view-shared anchors and anchor graphs which are identical for all views. Based on the work of [25], Chen et al. [27] presented a self-guided clustering model by using Global-guided-local strategy, which well takes into account local and

global information. However, the aforementioned methods overlook the impact of different dimensions of multi-view data, which weakens the representation ability of anchor graph [34], [35]. Wan et al. [35] proposed auto-weighted multi-view clustering (AWMVC) model. Although the above mentioned methods have achieved good performance in some scenarios, the efficiency of factorization decreases with high-dimensional data, as these methods typically apply NMF directly to the original data.

To address this limitation, Yang et al. [20] integrated anchor graphs within the SNMF/NMF framework and proposed an efficient multi-view clustering method with suitable constraints. However, they assumed that the indicator matrices of different views are completely identical, ignoring the difference between views in practical applications. Moreover, they can't well exploit the complementary information and spatial structure information embedded in across views due to the fact that the aforementioned methods leverage pixel by pixel error to learn the view consistent indicator matrix [15], [16], [36]. To solve the aforementioned problems, Li et al. [37] performed orthogonal non-negative matrix factorization on the anchor graphs to learn the view-consistent clustering indicator matrix with tensor Schatten p -norm minimization which well mines the complementary information embedded in views. But, like existing clustering methods, it does not consider the probability characteristics of the anchor graph when applying NMF to anchor graphs, and only use decision-level fusion to learn view consistent indicator matrix, ignoring data-level fusion which is important for clustering [21]. Inspired by the work of [21] and characteristics of anchor graphs, our proposed model leverages both data-level fusion and decision-level fusion to learn view-consistent indicator matrix which well exploits the complementary information between views.

3 NOTATIONS

In this section, we present the notations used throughout this paper, along with detailed explanations of the tensor operations employed. A comprehensive list of these notations and their respective meanings is provided in Table 1.

We proceed to introduce the tensor t-product and tensor Schatten p -norm as detailed in Definitions 1 and 2, respectively:

Definition 1 (t-product [38]). Let $\mathcal{A} \in \mathbb{R}^{n_1 \times m \times n_3}$ and $\mathcal{B} \in \mathbb{R}^{m \times n_2 \times n_3}$. The t-product $\mathcal{A} * \mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is defined as

$$\mathcal{A} * \mathcal{B} = \text{ifft}(\text{bdiag}(\overline{\mathcal{A}\mathcal{B}}), [], 3),$$

where $\overline{\mathcal{A}} = \text{bdiag}(\overline{\mathcal{A}})$ represents the block diagonal matrix, with its blocks being the frontal slices of $\overline{\mathcal{A}}$. $\text{ifft}(\text{bdiag}(\overline{\mathcal{A}\mathcal{B}}), [], 3)$ denotes performing the inverse Fast Fourier Transform on $\text{bdiag}(\overline{\mathcal{A}\mathcal{B}})$ along the third dimension.

Definition 2. [39] the tensor Schatten p -norm of $\mathcal{H} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is defined as:

$$\|\mathcal{H}\|_{\mathbb{S}} = \left(\sum_{i=1}^{n_2} \left\| \overline{\mathcal{H}}^{(i)} \right\|_{\mathbb{S}}^p \right)^{\frac{1}{p}} = \left(\sum_{i=1}^{n_2} \sum_{j=1}^h \sigma_j \left(\overline{\mathcal{H}}^{(i)} \right)^p \right)^{\frac{1}{p}}, \quad (1)$$

TABLE 1
Notations and Descriptions.

Notation	Descriptions
$\mathcal{H}$	3rd-order tensors
$\mathbf{H}$	matrices
$\mathbf{h}$	vectors
h_{ijk}	the entries of $\mathcal{H}$
$\mathbf{h}_{:jk}$	the column fiber of tensor $\mathcal{H}$
$\mathbf{h}_{i:k}$	the row fiber of tensor $\mathcal{H}$
$\mathbf{h}_{ij:}$	the tube fiber of tensor $\mathcal{H}$
$\mathbf{H}^{(i)}$	the i -th frontal slice of $\mathcal{H}$
$\overline{\mathcal{H}}$	the discrete Fourier transform (DFT) of $\mathcal{H}$ along the third dimension
$(\overline{\mathcal{H}}^{(v)})^H$	the Hermitian transpose of matrix $\overline{\mathcal{H}}^{(v)}$
$\text{tr}(\mathbf{H})$	the trace of matrix $\mathbf{H}$
$\ \mathcal{H}\ _F$	the F-norm of $\mathcal{H}$
$\text{fft}(\mathcal{H}, [], 3)$	performing the Fast Fourier Transform on $\mathcal{H}$ along the third dimension
$\text{ifft}(\mathcal{H}, [], 3)$	performing the inverse Fast Fourier Transform on $\mathcal{H}$ along the third dimension

where $0 < p \leq 1$, and $\sigma_j(\overline{\mathcal{H}}^{(i)})$ denotes the j -th singular value of $\overline{\mathcal{H}}^{(i)}$, and $h = \min(n_1, n_3)$,

It is noteworthy that for $0 < p \leq 1$, when p is carefully chosen, the Schatten p -norm can significantly enhance the approximation accuracy of the rank function, as discussed in [39]–[41].

Definition 3 (Tensor Fiber [42]). A tensor fiber refers to a 1D vector formed by fixing all other dimensions of a tensor, serving as the 1D basic component of the tensor. A third-order tensor has column fibers, row fibers, and tube fibers, denoted as $\mathbf{h}_{:jk}$, $\mathbf{h}_{i:k}$, and $\mathbf{h}_{ij:}$, respectively.

Definition 4 (Tensor Slice [42]). A tensor slice refers to a matrix formed by fixing all other dimensions of a tensor, with the remaining two dimensions. A third-order tensor has horizontal slices, lateral slices, and frontal slices, denoted as $\mathbf{H}_{i::}$, $\mathbf{H}_{:j:}$, and $\mathbf{H}_{::k}$ (or $\mathbf{H}^{(k)}$), respectively.

Definition 5 (t-SVD [43]). Given the tensor $\mathcal{Z} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the tensor Singular Value Decomposition (t-SVD) is given as $\mathcal{Z} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$, where $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensors, $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is the f-diagonal tensor.

4 METHODOLOGY

4.1 Motivation and Objective

Anchor graph clustering based on NMF has become a hot research topic due to its interpretability and effectiveness in processing large-scale data. Building upon this paradigm, a number of NMF-based multi-view clustering methods have been proposed, aiming to exploit complementary information from multiple views via anchor graphs. Some methods introduce shared consensus representations, such as a shared anchor graph matrix (e.g., [44]), while others impose regularization terms to encourage agreement among view-specific factors.

In general, these approaches can be summarized as:

$$\min_{\mathbf{H}^{(v)}, \mathbf{G}^{(v)}} \sum_{v=1}^V \left\| \mathbf{S}^{(v)} - \mathbf{H}^{(v)} \mathbf{G}^{(v)T} \right\|_F^2 + \mathcal{R}(\mathbf{H}^{(1)}, \dots, \mathbf{H}^{(V)}) \quad (2)$$

$$\text{s.t. } \mathbf{H}^{(v)} \geq 0$$

where $\mathbf{S}^{(v)} \in \mathbb{R}^{n \times m}$ is the anchor graph of the v -th view, n is the number of samples, m is the number of anchors. $\mathbf{H}^{(v)}$ and $\mathbf{G}^{(v)}$ denote the embedding and basis matrix of $\mathbf{S}^{(v)}$, respectively. V denotes the number of views.

Although consensus-based strategies effectively enhance alignment across views, the strong constraints may reduce modeling flexibility when views exhibit heterogeneous or view-specific structures. This motivates us to design a more adaptive mechanism to balance consensus and diversity across views.

In Eq. (2), the first term is non-negative matrix factorization, and the second term is regularization. It can be seen that existing methods apply NMF to obtain embedding, i.e., sample indicator matrices for each view independently. The indicator matrices of views are then fused at the decision level to generate a view-consistent label matrix (See Fig. 1). Thus, a major limitation of these approaches is that they focus solely on decision-level fusion, overlooking the implicit relationships between views at the data level, which helps to enhance the representation ability of coefficient matrix. Moreover, the existing methods rely on the relationship between NMF and K-means to derive soft labels, ignoring the probability characteristics of anchor graph.

In order to effectively utilize complementary information between views to improve the stability of coefficient matrices and basis matrices, we extend the classical non-negative matrix factorization to non-negative tensor factorization (NTF) and apply NTF to the third-order tensorized anchor graph. We have

$$\min_{\mathcal{H}, \mathcal{G}} \left\| \mathcal{S} - \mathcal{H} * \mathcal{G}^T \right\|_F^2 + \mathcal{R}(\mathcal{H}) \quad (3)$$

$$\text{s.t. } \mathcal{H} \geq 0, \mathcal{G} \geq 0$$

where, tensorized anchor graph $\mathcal{S} \in \mathbb{R}^{n \times m \times V}$ is composed of anchor graphs of views (See Fig. 2), $\mathcal{H} \in \mathbb{R}^{n \times c \times V}$ and $\mathcal{G} \in \mathbb{R}^{m \times c \times V}$ denotes the basis tensor and coefficient tensor, respectively. $\mathcal{R}(\mathcal{H})$ is regularization term.

In Eq. (3), the first term factorizes tensorized anchor graph $\mathcal{S}$ into the product of tensor $\mathcal{H}$ and tensor $\mathcal{G}$. Thus, the representation of each view is directly related to the other views, achieving the data-level fusion. But, Eq. (3) can't exploit the probability characteristic of anchor graph.

Given an anchor graph $\mathbf{S}$, the element S_{ij} represents the similarity between the i -th sample $\mathbf{x}_i$ and the j -th anchor $\mathbf{a}_j$, i.e., probability [23]. According to the transition probability, we have

$$s_{ij} = p(\mathbf{a}_j | \mathbf{x}_i) = \sum_{k=1}^C p(c_k | \mathbf{x}_i) p(\mathbf{a}_j | c_k) \quad (4)$$

$$= \sum_{k=1}^C H_{ik} G_{kj} = \langle H_{i:,}, G_{j:,} \rangle$$

Then,

$$\mathbf{S} = \mathbf{H} \mathbf{G}^T \quad (5)$$

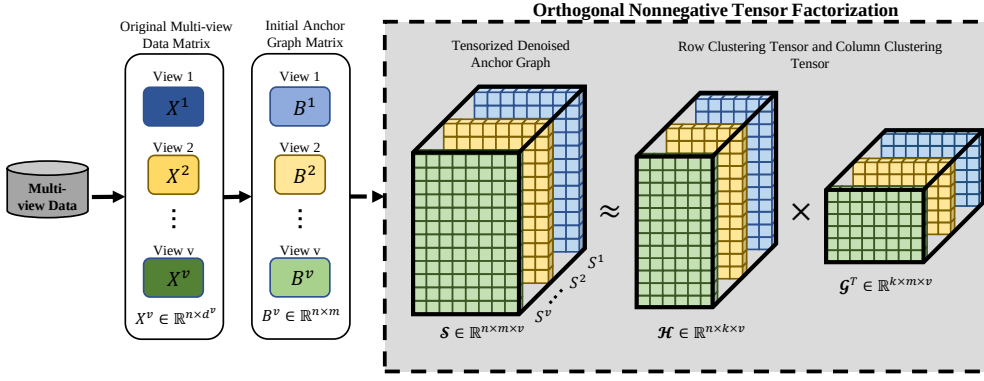


Fig. 2. Process flow of our model. We first construct the initial anchor graphs $\{\mathbf{B}^{(v)}\}_{v=1}^V$ as a tensor. Then, we incorporate the learning of high-quality anchor graph $\mathbf{S}$ and tensor factorization into a unified framework so that they can complement each other.

where $\mathbf{H} \in \mathbb{R}^{n \times c}$ and $\mathbf{G} \in \mathbb{R}^{m \times c}$ denote sample indicator matrix and anchor indicator matrix, respectively. So, the corresponding constraints are $\mathbf{H} \geq 0, \mathbf{H} \cdot \mathbf{1} = \mathbf{1}, \mathbf{G} \geq 0, \mathbf{G} \cdot \mathbf{1} = \mathbf{1}$. It is easy to prove that these constraints lead to the trivial solution of Eq. (5), that is, all samples are divided into a cluster. To avoid this problem and inspired by [31], we modify the constraint on $\mathbf{H}$ as $\mathbf{H} \geq 0, \mathbf{H}^T \mathbf{H} = \mathbf{I}$. The position of the largest value in the row indicates the cluster to which the sample belongs.

In practical applications, anchor graphs are often inevitably contaminated by noise, which may adversely affect subsequent representation learning and clustering performance. To address this issue, we introduce a denoising mechanism by decomposing the constructed anchor graph $\mathbf{B}$ into a cleaner anchor graph $\mathbf{S}$ and a noise term $\mathbf{E}$. This leads to the following optimization problem:

$$\min_{\mathbf{S}, \mathbf{E}} \|\mathbf{S}\|_* + \|\mathbf{E}\|_1, \text{ s.t. } \mathbf{B} = \mathbf{S} + \mathbf{E}. \quad (6)$$

According to Eq. (5), the clean tensorized anchor graph can be factorized as $\mathbf{S} = \mathbf{H} * \mathbf{G}^T$, from which it follows that

$$\text{rank}(\mathbf{S}) \leq \min\{\text{rank}(\mathbf{H}), \text{rank}(\mathbf{G})\}. \quad (7)$$

Since the ranks of $\mathbf{H}$ and $\mathbf{G}$ are both bounded by the number of clusters c , $\mathbf{S}$ naturally exhibits a low-rank structure. Therefore, the low-rank property of $\mathbf{S}$ is implicitly guaranteed by the factorization, and the nuclear norm regularization in the above formulation can be safely omitted.

Consequently, the optimization problem can be equivalently simplified as:

$$\begin{aligned} \min_{\mathbf{E}, \mathbf{H}, \mathbf{G}} \|\mathbf{E}\|_1, \\ \text{s.t. } \mathbf{B} = \mathbf{H} * \mathbf{G}^T + \mathbf{E}, \mathbf{H} \geq 0, \mathbf{H}^T * \mathbf{H} = \mathbf{I}, \\ \mathbf{G}^{(v)} \geq 0, \mathbf{G}^{(v)} \cdot \mathbf{1} = \mathbf{1}. \end{aligned} \quad (8)$$

This formulation enables effective suppression of sparse noise in the anchor graph while preserving the expressive power of the original NTF-based model, thereby yielding a more robust anchor graph representation.

In order to better explore the complementary information between views, inspired by the work of [14], [15], we leverage the tensor Schatten p -norm minimization on $\mathbf{H}$ and $\mathbf{G}$ to learn view-consistent samples and anchors indicator matrices (See Fig. 2 and Fig. 3). Details on the tensor Schatten p -norm

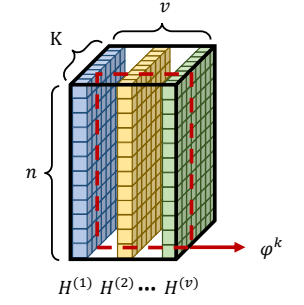


Fig. 3. Interpretation of the tensor Schatten p -norm

are provided in Remark 1. This plays a role in decision level fusion. Thus, our final objective function is

$$\begin{aligned} \min_{\mathbf{E}, \mathbf{H}, \mathbf{G}} \|\mathbf{E}\|_1 + \lambda_1 \|\mathbf{H}\|_{\mathbb{S}^p}^p + \lambda_2 \|\mathbf{G}\|_{\mathbb{S}^p}^p \\ \text{s.t. } \mathbf{B} = \mathbf{H} * \mathbf{G}^T + \mathbf{E}, \mathbf{H} \geq 0, \mathbf{H}^T * \mathbf{H} = \mathbf{I}, \\ \mathbf{G}^{(v)} \geq 0, \mathbf{G}^{(v)} \cdot \mathbf{1} = \mathbf{1}, \end{aligned} \quad (9)$$

where $0 < p \leq 1$, and λ_1, λ_2 are the hyperparameters. $\mathbf{B} \in \mathbb{R}^{n \times m \times V}$ is the tensorized anchor graph. $\mathbf{H} \in \mathbb{R}^{n \times c \times V}$ represents the clustering indicator tensor of the samples. $\mathbf{E} \in \mathbb{R}^{n \times m \times V}$ represents the noise in the initial anchor graph. The matrix $\mathbf{G}^{(v)}$ represents the v -th frontal slice of the tensor $\mathbf{G}$, as illustrated in Fig 2. Figure 2 shows the workflow of our proposed model.

Remark 1 (Explanation of the tensor Schatten p -norm).

For a matrix $\mathbf{A}$ with singular values $\sigma_1, \dots, \sigma_h$ sorted in descending order, the Schatten p -norm is defined as $\|\mathbf{A}\|_{\mathbb{S}^p}^p = \sigma_1^p + \dots + \sigma_h^p$. As p approaches 0, $\lim_{p \rightarrow 0} \|\mathbf{A}\|_{\mathbb{S}^p}^p$ equals the rank of $\mathbf{A}$, which is the number of non-zero singular values. Thus, Schatten p -norm minimization for $p < 1$ often achieves a closer approximation to the desired rank than the nuclear norm ($p = 1$).

For the tensor $\mathbf{H}$, as depicted in Fig 3, its k -th lateral slice Φ^k represents the relationship between n samples and the k -th cluster across different views. Multi-view clustering aims to harmonize the sample-cluster relationships across views, aligning $\mathbf{H}_{:,k}^1, \dots, \mathbf{H}_{:,k}^v$. However, clustering structures often vary across views. Applying the tensor Schatten p -norm to $\mathbf{H}$ ensures that Φ^k maintains a spatially low-rank structure, leveraging complementary information across views and promoting clustering consistency.

4.2 Optimization

Since directly solving Eq. (9) is challenging, we adopt the Augmented Lagrange Multiplier (ALM) method for optimization. Specifically, we introduce three auxiliary variables $\mathbf{Q}$, $\mathbf{J}$, and $\mathbf{F}$, and enforce the constraints $\mathbf{H} = \mathbf{Q}$, $\mathbf{H} = \mathbf{J}$, and $\mathbf{G} = \mathbf{F}$, respectively, with the additional condition $\mathbf{Q} \geq 0$. Consequently, the objective function (9) can be reformulated into an unconstrained format as follows:

$$\begin{aligned} \min & \|\mathcal{E}\|_1 + \lambda_1 \|\mathcal{J}\|_{\mathbb{S}}^p + \lambda_2 \|\mathcal{F}\|_{\mathbb{S}}^p + \frac{\mu}{2} \left\| \mathcal{H} - \mathcal{Q} + \frac{\mathcal{Y}_1}{\mu} \right\|_F^2 \\ & + \frac{\rho}{2} \left\| \mathcal{H} - \mathcal{J} + \frac{\mathcal{Y}_2}{\rho} \right\|_F^2 + \frac{\sigma}{2} \left\| \mathcal{G} - \mathcal{F} + \frac{\mathcal{Y}_3}{\sigma} \right\|_F^2 \\ & + \frac{\beta}{2} \left\| \mathcal{B} - \mathcal{H} * \mathcal{G}^T - \mathcal{E} + \frac{\mathcal{Y}_4}{\beta} \right\|_F^2, \\ \text{s.t. } & \mathcal{Q} \geq 0, \mathcal{H}^T * \mathcal{H} = \mathcal{I}, \mathcal{G}^{(v)} \geq 0, \mathcal{G}^{(v)} \cdot \mathbf{1} = \mathbf{1}, \end{aligned} \quad (10)$$

here, $\mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Y}_3$, and $\mathcal{Y}_4$ denote the Lagrange multipliers, while μ, ρ, σ , and β represent the penalty parameters. To address this optimization challenge (10), we employ an alternating optimization approach. Each variable is optimized independently while keeping the others fixed. The optimization process comprises six steps:

• **Solving for $\mathcal{G}$ with fixed $\mathcal{E}, \mathcal{Q}, \mathcal{H}, \mathcal{J}$, and $\mathcal{F}$.** The objective function (10) transforms into:

$$\begin{aligned} \min & \frac{\sigma}{2} \left\| \mathcal{G} - \mathcal{F} + \frac{\mathcal{Y}_3}{\sigma} \right\|_F^2 + \frac{\beta}{2} \left\| \mathcal{B} - \mathcal{H} * \mathcal{G}^T - \mathcal{E} + \frac{\mathcal{Y}_4}{\beta} \right\|_F^2, \\ \text{s.t. } & \mathcal{G}^{(v)} \geq 0, \mathcal{G}^{(v)} \cdot \mathbf{1} = \mathbf{1} \end{aligned} \quad (11)$$

Further, (11) can be reformulated in the frequency domain as:

$$\begin{aligned} \min_{\mathcal{G}^{(v)} \geq 0, \mathcal{G}^{(v)} \cdot \mathbf{1} = \mathbf{1}} & \sum_{v=1}^V \frac{\sigma}{2} \left\| \bar{\mathcal{G}}^{(v)} - \bar{\mathcal{F}}^{(v)} + \frac{\bar{\mathcal{Y}}_3^{(v)}}{\sigma} \right\|_F^2 \\ & + \sum_{v=1}^V \frac{\beta}{2} \left\| \bar{\mathcal{B}}^{(v)} - \bar{\mathcal{H}}^{(v)} (\bar{\mathcal{G}}^{(v)})^H - \bar{\mathcal{E}}^{(v)} + \frac{\bar{\mathcal{Y}}_4^{(v)}}{\beta} \right\|_F^2, \end{aligned} \quad (12)$$

where $\bar{\mathcal{G}} = \text{fft}(\mathcal{G}, [], 3)$, and other variables are transformed similarly. $(\bar{\mathcal{G}}^{(v)})^H$ is the Hermitian transpose of matrix $\bar{\mathcal{G}}^{(v)}$.

Clearly, the equation (12) can be reformulated as:

$$\begin{aligned} \min_{\mathcal{G}^{(v)} \geq 0, \mathcal{G}^{(v)} \cdot \mathbf{1} = \mathbf{1}} & \sum_{v=1}^V \left\{ \frac{\sigma + \beta}{2} \text{tr}[(\bar{\mathcal{G}}^{(v)})^H \bar{\mathcal{G}}^{(v)}] \right. \\ & \left. - \beta \text{tr}[(\bar{\mathcal{G}}^{(v)})^H (\bar{\mathcal{A}}_1^{(v)})^H \bar{\mathcal{H}}^{(v)}] - \sigma \text{tr}[(\bar{\mathcal{G}}^{(v)})^H (\bar{\mathcal{F}}^{(v)} - \frac{\bar{\mathcal{Y}}_3^{(v)}}{\sigma})] \right\}, \end{aligned} \quad (13)$$

where $\bar{\mathcal{A}}_1^{(v)} = \bar{\mathcal{B}}^{(v)} - \bar{\mathcal{E}}^{(v)} + \frac{\bar{\mathcal{Y}}_4^{(v)}}{\beta}$.

Further simplification of (13) yields:

$$\begin{aligned} \min & \sum_{v=1}^V \frac{\sigma + \beta}{2} \left\{ \text{tr}[(\bar{\mathcal{G}}^{(v)})^H \bar{\mathcal{G}}^{(v)}] - 2 \text{tr}[(\bar{\mathcal{G}}^{(v)})^H \bar{\mathcal{A}}_2^{(v)}] \right\} \\ \text{s.t. } & \mathcal{G}^{(v)} \geq 0, \mathcal{G}^{(v)} \cdot \mathbf{1} = \mathbf{1}, \end{aligned} \quad (14)$$

where $\bar{\mathcal{A}}_2^{(v)} = [\beta (\bar{\mathcal{A}}_1^{(v)})^H \bar{\mathcal{H}}^{(v)} + \sigma \bar{\mathcal{F}}^{(v)} - \bar{\mathcal{Y}}_3^{(v)}] / (\sigma + \beta)$.

By returning (14) to the time domain and simplifying it, we can get:

$$\min_{\mathcal{G}^{(v)} \geq 0, \mathcal{G}^{(v)} \cdot \mathbf{1} = \mathbf{1}} \frac{\sigma + \beta}{2} \|\mathcal{G} - \mathcal{A}_2\|_F^2 \quad (15)$$

We can derive the solution of (15) as $\mathcal{G}_i^{(v)} = (\mathcal{A}_2)_i^{(v)} + \gamma \mathbf{1}$ according to [45], where $\mathcal{G}_i^{(v)}$ represents pulling the matrix $\mathcal{G}^{(v)}$ into a vector and γ is a Lagrange multiplier.

• **Solve for $\mathcal{H}$ with fixed $\mathcal{E}, \mathcal{Q}, \mathcal{G}, \mathcal{J}$ and $\mathcal{F}$.** (10) then transforms into:

$$\begin{aligned} \min_{\mathcal{H}^T * \mathcal{H} = \mathcal{I}} & \frac{\mu}{2} \left\| \mathcal{H} - \mathcal{Q} + \frac{\mathcal{Y}_1}{\mu} \right\|_F^2 + \frac{\rho}{2} \left\| \mathcal{H} - \mathcal{J} + \frac{\mathcal{Y}_2}{\rho} \right\|_F^2 \\ & + \frac{\beta}{2} \left\| \mathcal{B} - \mathcal{H} * \mathcal{G}^T - \mathcal{E} + \frac{\mathcal{Y}_4}{\beta} \right\|_F^2 \end{aligned} \quad (16)$$

Further, (16) can be reformulated in the frequency domain as:

$$\begin{aligned} \min_{(\bar{\mathcal{H}}^{(v)})^H \bar{\mathcal{H}}^{(v)} = \mathcal{I}} & \sum_{v=1}^V \frac{\mu}{2} \left\| \bar{\mathcal{H}}^{(v)} - \bar{\mathcal{Q}}^{(v)} + \frac{\bar{\mathcal{Y}}_1^{(v)}}{\mu} \right\|_F^2 \\ & + \sum_{v=1}^V \frac{\rho}{2} \left\| \bar{\mathcal{H}}^{(v)} - \bar{\mathcal{J}}^{(v)} + \frac{\bar{\mathcal{Y}}_2^{(v)}}{\rho} \right\|_F^2 \\ & + \sum_{v=1}^V \frac{\beta}{2} \left\| \bar{\mathcal{B}}^{(v)} - \bar{\mathcal{H}}^{(v)} (\bar{\mathcal{G}}^{(v)})^H - \bar{\mathcal{E}}^{(v)} + \frac{\bar{\mathcal{Y}}_4^{(v)}}{\beta} \right\|_F^2, \end{aligned} \quad (17)$$

Clearly, the equation (17) can be reformulated as:

$$\begin{aligned} \min_{(\bar{\mathcal{H}}^{(v)})^H \bar{\mathcal{H}}^{(v)} = \mathcal{I}} & \sum_{v=1}^V \left\{ -\beta \text{tr}[(\bar{\mathcal{H}}^{(v)})^H \bar{\mathcal{A}}_1^{(v)} \bar{\mathcal{G}}^{(v)}] \right. \\ & \left. - \mu \text{tr}[(\bar{\mathcal{H}}^{(v)})^H (\bar{\mathcal{Q}}^{(v)} - \frac{\bar{\mathcal{Y}}_1^{(v)}}{\mu})] - \rho \text{tr}[(\bar{\mathcal{H}}^{(v)})^H (\bar{\mathcal{J}}^{(v)} - \frac{\bar{\mathcal{Y}}_2^{(v)}}{\rho})] \right\}, \end{aligned} \quad (18)$$

where $\bar{\mathcal{A}}_1^{(v)} = \bar{\mathcal{B}}^{(v)} - \bar{\mathcal{E}}^{(v)} + \frac{\bar{\mathcal{Y}}_4^{(v)}}{\beta}$.

Further simplification of (18) yields:

$$\max_{(\bar{\mathcal{H}}^{(v)})^H \bar{\mathcal{H}}^{(v)} = \mathcal{I}} \sum_{v=1}^V \text{tr}[(\bar{\mathcal{H}}^{(v)})^H \bar{\mathcal{A}}_3^{(v)}] \quad (19)$$

where $\bar{\mathcal{A}}_3^{(v)} = \beta \bar{\mathcal{A}}_1^{(v)} \bar{\mathcal{G}}^{(v)} + \mu \bar{\mathcal{Q}}^{(v)} - \bar{\mathcal{Y}}_1^{(v)} + \rho \bar{\mathcal{J}}^{(v)} - \bar{\mathcal{Y}}_2^{(v)}$.

To address (19), we introduce the following theorems:

Theorem 1. Given two matrices $\mathbf{P} \in \mathbb{R}^{n_1 \times n_2}$, $\mathbf{Q} \in \mathbb{R}^{n_1 \times n_2}$, we have

$$\text{tr}(\mathbf{P}^H \mathbf{Q}) \leq \|\mathbf{P}\|_F \|\mathbf{Q}\|_F \quad (20)$$

Equality holds if and only if $\mathbf{Q}$ is a multiple of $\mathbf{P}$.

Proof 1. $\text{vec}(\mathbf{P})$ and $\text{vec}(\mathbf{Q})$ denote the column vectors that concatenate all the columns of matrix $\mathbf{P}$ and $\mathbf{Q}$, respectively. Then,

$$\text{tr}(\mathbf{P}^H \mathbf{Q}) = (\text{vec}(\mathbf{P}))^H \text{vec}(\mathbf{Q}). \quad (21)$$

Leveraging Cauchy-Schwarz inequality, we have

$$\begin{aligned} (\text{vec}(\mathbf{P}))^H \text{vec}(\mathbf{Q}) & \leq \|\text{vec}(\mathbf{P})\|_2 \|\text{vec}(\mathbf{Q})\|_2 \\ & = \|\mathbf{P}\|_F \|\mathbf{Q}\|_F. \end{aligned} \quad (22)$$

Comparing inequalities (21) and (22) yields

$$\text{tr}(\mathbf{P}^H \mathbf{Q}) \leq \|\mathbf{P}\|_F \|\mathbf{Q}\|_F \quad (23)$$

Theorem 2. Given two matrices $\mathbf{P} \in \mathbb{R}^{n_1 \times n_2}$, $\mathbf{Q} \in \mathbb{R}^{n_1 \times n_2}$, then the optimal solution of

$$\arg \max_{\mathbf{P}^H \mathbf{P} = \mathbf{I}_{n_2}} \text{tr}(\mathbf{P}^H \mathbf{Q}) \quad (24)$$

is $\mathbf{P} = \mathbf{U} \mathbf{W}^H$. Where, $\mathbf{Q} = \mathbf{U} \mathbf{\Sigma} \mathbf{W}^H$ is the singular value decomposition (SVD) of $\mathbf{Q}$, $\mathbf{U} \in \mathbb{R}^{n_1 \times n_2}$, $\mathbf{W} \in \mathbb{R}^{n_2 \times n_2}$ is an orthogonal matrix, $\mathbf{\Sigma} \in \mathbb{R}^{n_2 \times n_2}$, $\mathbf{U}^H \mathbf{U} = \mathbf{I}_{n_2}$.

Proof 2.

$$\begin{aligned} \text{tr}(\mathbf{P}^H \mathbf{Q}) &= \text{tr}(\mathbf{P}^H \mathbf{U} \Sigma \mathbf{W}^H) \\ &= \text{tr}(\mathbf{U} \Sigma^{1/2} \Sigma^{1/2} \mathbf{W}^H \mathbf{P}^H). \end{aligned} \quad (25)$$

Using Theorem 1, we get

$$\begin{aligned} \text{tr}(\mathbf{P}^H \mathbf{Q}) &\leq \|\mathbf{U} \Sigma^{1/2}\|_F \|\Sigma^{1/2} \mathbf{W}^H \mathbf{P}^H\|_F \\ &= \|\Sigma^{1/2}\|_F \|\Sigma^{1/2}\|_F. \end{aligned} \quad (26)$$

Equality holds only when

$$\Sigma^{1/2} \mathbf{U}^H = \Sigma^{1/2} \mathbf{W}^H \mathbf{P}^H \quad (27)$$

Thus, $\mathbf{P} = \mathbf{U} \mathbf{W}^H$.

By Theorem 2, we can derive the solution of (19) as:

$$\overline{\mathcal{H}}^{(v)} = \overline{\Lambda}_2^{(v)} (\overline{\mathcal{V}}_2^{(v)})^H, \quad (28)$$

where $\overline{\Lambda}_2^{(v)}$ and $\overline{\mathcal{V}}_2^{(v)}$ are determined through SVD of $\overline{\mathcal{A}}_3^{(v)}$.

• **Solve $\mathcal{Q}$ with fixed $\mathcal{E}, \mathcal{G}, \mathcal{H}, \mathcal{J}$ and $\mathcal{F}$.** (10) becomes:

$$\min_{\mathcal{Q} \geq 0} \frac{\mu}{2} \left\| \mathcal{H} - \mathcal{Q} + \frac{\mathcal{Y}_1}{\mu} \right\|_F^2 \quad (29)$$

Transforming this into the frequency domain and rearranging the variables, we obtain:

$$\min_{\mathcal{Q}^{(v)} \geq 0} \sum_{v=1}^V \frac{\mu}{2} \left\| \overline{\mathcal{Q}}^{(v)} - (\overline{\mathcal{H}}^{(v)} + \frac{\overline{\mathcal{Y}}_1^{(v)}}{\mu}) \right\|_F^2 \quad (30)$$

The solution for Equation (30) is given by:

$$\overline{\mathcal{Q}}^{(v)} = \left(\overline{\mathcal{H}}^{(v)} + \frac{\overline{\mathcal{Y}}_1^{(v)}}{\mu} \right)_+ \quad (31)$$

• **Solve for $\mathcal{J}$ with fixed $\mathcal{E}, \mathcal{Q}, \mathcal{H}, \mathcal{G}$, and $\mathcal{F}$.** The equation (10) is reformulated as:

$$\min \lambda_1 \|\mathcal{J}\|_{\mathbb{S}}^p + \frac{\rho}{2} \left\| \mathcal{H} - \mathcal{J} + \frac{\mathcal{Y}_2}{\rho} \right\|_F^2, \quad (32)$$

where, after completing the square with respect to $\mathcal{J}$, we obtain

$$\mathcal{J}^* = \arg \min \frac{1}{2} \left\| \mathcal{J} - (\mathcal{H} + \frac{\mathcal{Y}_2}{\rho}) \right\|_F^2 + \frac{\lambda_1}{\rho} \|\mathcal{J}\|_{\mathbb{S}}^p, \quad (33)$$

yielding a closed-form solution as outlined in Lemma 1 [39]:

Lemma 1. Given the input $\mathcal{Z} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ with a t-SVD $\mathcal{Z} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$, the optimal solution for

$$\min_{\mathcal{X}} \frac{1}{2} \|\mathcal{X} - \mathcal{Z}\|_F^2 + \tau \|\mathcal{X}\|_{\mathbb{S}}^p \quad (34)$$

is $\mathcal{X}^* = \Gamma_{\tau}(\mathcal{Z}) = \mathcal{U} * \text{ifft}(P_{\tau}(\overline{\mathcal{Z}})) * \mathcal{V}^T$, where τ is the input parameter, $\tau > 0$, $P_{\tau}(\overline{\mathcal{Z}}) = \text{diag}(z_1, z_2, \dots, z_l)$ is an f-diagonal 3rd-order tensor. Its diagonal elements $z_1, z_2, \dots, z_l$ ($l = \min(n_1, n_2)$) can be determined using the GST algorithm described in [39].

Thus, the solution for (33) is expressed as:

$$\mathcal{J}^* = \Gamma_{\frac{\lambda_1}{\rho}}(\mathcal{H} + \frac{\mathcal{Y}_2}{\rho}). \quad (35)$$

• **Solve for $\mathcal{F}$ with fixed $\mathcal{E}, \mathcal{Q}, \mathcal{H}, \mathcal{J}$ and $\mathcal{G}$.** The objective function (10) simplifies to:

$$\min \lambda_2 \|\mathcal{F}\|_{\mathbb{S}}^p + \frac{\sigma}{2} \left\| \mathcal{G} - \mathcal{F} + \frac{\mathcal{Y}_3}{\sigma} \right\|_F^2, \quad (36)$$

By completing the square with respect to $\mathcal{F}$, we obtain:

$$\mathcal{F}^* = \arg \min \frac{1}{2} \left\| \mathcal{F} - (\mathcal{G} + \frac{\mathcal{Y}_3}{\sigma}) \right\|_F^2 + \frac{\lambda_2}{\sigma} \|\mathcal{F}\|_{\mathbb{S}}^p, \quad (37)$$

Applying Lemma 1, the solution to (37) is given by:

$$\mathcal{F}^* = \Gamma_{\frac{\lambda_2}{\sigma}}(\mathcal{G} + \frac{\mathcal{Y}_3}{\sigma}). \quad (38)$$

• **Solving for $\mathcal{E}$ with fixed $\mathcal{G}, \mathcal{Q}, \mathcal{H}, \mathcal{J}$, and $\mathcal{F}$.** The objective function (10) transforms into:

$$\min \|\mathcal{E}\|_1 + \frac{\beta}{2} \left\| \mathcal{B} - \mathcal{H} * \mathcal{G}^T - \mathcal{E} + \frac{\mathcal{Y}_4}{\beta} \right\|_F^2 \quad (39)$$

The optimal solution of Formula (39) with respect to the variable $\mathcal{E}$ is:

$$\mathcal{E}^* = \arg \min_{\mathcal{E}} \frac{1}{\beta} \|\mathcal{E}\|_1 + \frac{1}{2} \|\mathcal{E} - \mathcal{A}_4\|_F^2 \quad (40)$$

where $\mathcal{A}_4 = \mathcal{B} - \mathcal{H} * \mathcal{G}^T + \frac{\mathcal{Y}_4}{\beta}$. The optimal solution of Eq.(40) is $\mathbb{S}_{\frac{1}{\beta}}[\mathcal{A}_4]$, where $\mathbb{S}_{\frac{1}{\beta}}[a] = \text{sign}(a) = \max(|a| - \frac{1}{\beta}, 0)$ is the soft-thresholding operator [46].

Algorithm 1: Non-negative Tensor Factorization based Bi-Clustering on Anchor Graphs

Require: Data matrices $\{\mathbf{X}^{(v)}\}_{v=1}^V \in \mathbb{R}^{N \times d_v}$; anchors numbers m ; cluster number K .

Ensure: Cluster labels $\mathbf{Y}$ of each data points.

- 1: **Initialize:** $\mu = 10^{-5}$, $\rho = 10^{-5}$, $\sigma = 10^{-5}$, $\beta = 10^{-5}$, $\eta = 1.2$, $\mathcal{Y}_1 = 0$, $\mathcal{Y}_2 = 0$, $\mathcal{Y}_3 = 0$, $\mathcal{Y}_4 = 0$ and $\overline{\mathcal{H}}^{(v)}$ is identity matrix;
 - 2: Compute graph matrix $\mathbf{B}^{(v)}$ of each views and construct $\mathbf{B}^{(v)}$ as a tensor $\mathcal{B}$;
 - 3: **while** not condition **do**
 - 4: Update $\mathcal{G}$ by solving (15);
 - 5: Update $\overline{\mathcal{H}}^{(v)}$ by solving (28);
 - 6: Update $\overline{\mathcal{Q}}^{(v)}$ by solving (31);
 - 7: Update $\mathcal{J}$ by using (35);
 - 8: Update $\mathcal{F}$ by using (38);
 - 9: Update $\mathcal{E}$ by solving (40);
 - 10: Update $\mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Y}_3, \mathcal{Y}_4, \mu, \rho, \sigma$ and β :
 $\mathcal{Y}_1 = \mathcal{Y}_1 + \mu(\mathcal{H} - \mathcal{Q})$, $\mathcal{Y}_2 = \mathcal{Y}_2 + \rho(\mathcal{H} - \mathcal{J})$,
 $\mathcal{Y}_3 = \mathcal{Y}_3 + \sigma(\mathcal{G} - \mathcal{F})$,
 $\mathcal{Y}_4 = \mathcal{Y}_4 + \beta(\mathcal{B} - \mathcal{H} * \mathcal{G}^T - \mathcal{E})$, $\mu = \min(\eta\mu, 10^{13})$,
 $\rho = \min(\eta\rho, 10^{13})$, $\sigma = \min(\eta\sigma, 10^{13})$,
 $\beta = \min(\eta\beta, 10^{13})$;
 - 11: **end while**
 - 12: Calculate the K clusters by using
 $\mathbf{H} = \sum_{v=1}^V \mathbf{H}^{(v)} / V$;
 - 13: **return** Clustering result.
-

4.3 Computational Complexity and Scalability Analysis

In this section, we analyze the computational complexity and discuss the scalability of the proposed method. We define the following notations: $d^{(v)}$ represents the dimension of the v -th view, V is the total number of views, n denotes the number of samples, and c stands for the number of clusters.

Let $h = \sum_{v=1}^V d^{(v)}$ be the aggregate dimension across all views. The complexity required for constructing the anchor

TABLE 2
Multi-view datasets used in our experiments

#Dataset	#Samples	#View	#Class	#Feature
MSRC	210	5	7	24, 576, 512, 256, 254
ORL	400	3	40	4096, 3304, 6750
HandWritten4	2000	4	10	76, 216, 47, 6
Mnist4	4000	3	4	30, 9, 30
Scene15	4485	3	15	1800, 1180, 1240
Reuters	18758	5	6	21531, 24892, 34251, 15506, 11547
NUS	30000	5	31	64, 225, 144, 73, 128
AWA	30475	6	50	2688, 2000, 252, 2000, 2000, 2000
Noisy MNIST	50000	2	10	784, 784
YoutubeFace	101499	5	31	64, 512, 64, 647, 838

graph $\mathcal{B}$ is $\mathcal{O}(nmhV + nmV \log m)$. For the update procedures, the computational cost for updating $\mathcal{G}$ is $\mathcal{O}(mc \log c)$. Updating $\mathcal{E}$ incurs a cost of $\mathcal{O}(nmCV + nmV \log(m))$. Solving $\mathcal{H}$ requires $\mathcal{O}(nVc^2 + mcV \log(mV))$, $\mathcal{Q}$ requires $\mathcal{O}(nVc)$, $\mathcal{J}$ requires $\mathcal{O}(2ncV \log(cV) + ncV^2)$, and $\mathcal{F}$ requires $\mathcal{O}(2mcV \log(cV) + mcV^2)$.

Given that $n \gg c$, $n \gg V$, and $n > m$, the overall complexity is approximately $\mathcal{O}(n)$. This demonstrates a linear relationship with respect to the number of samples. Consequently, our algorithm achieves linear time complexity in both finding the optimal solution and subsequent processing steps, ensuring efficient performance in clustering tasks.

The scalability of the proposed method is primarily governed by the number of anchors m , which is typically set as a small fraction of n . By controlling m , the size of the anchor graph tensor and the associated tensor operations remain manageable even when n increases significantly. Compared with sample-level tensor clustering methods, which often suffer from quadratic or cubic complexity in n , the proposed anchor-based tensor factorization framework provides a more scalable solution for large-scale multi-view clustering.

5 EXPERIMENTS

In this section, we conduct comprehensive comparative experiments to evaluate the proposed methods. The effectiveness of clustering is measured using three key metrics: Accuracy (ACC), Normalized Mutual Information (NMI), and Purity. Higher values for these metrics correspond to better clustering performance. To ensure robustness, we performed 20 independent trials for each method and recorded the mean of the results. The datasets employed in our experiments are detailed in Table 2. We compare our approach against the following state-of-the-art methods: MultiNMF [33], NMF-CC [47], GMC [48], LMVSC [26], PLCMF [24], FPMVSCAG [25], FMCNOF [20], SFMC [29], AWMVC [35], FSMSC [27], TBGL [15], Orth-NTF [21], LMVC-MFCG [49], EMVC-NTLC [37], AGFS-OMVC [12], ALPC [50], Dual MoE MVKM [51], and LMTC [52]. Experiments using the MSRC, ORL, HandWritten4, Mnist4, and Scene15 datasets were conducted on a laptop equipped with an Intel Core i5-8300H CPU and 16 GB RAM, utilizing Matlab R2021b. In contrast, the Reuters, NUS, AWA, NoisyMnist and YoutubeFace datasets were processed on a standard Windows 10 server, featuring dual Intel(R) Xeon(R) Gold 6230 CPUs at 2.10 GHz and 128 GB RAM, with MATLAB R2020b.

5.1 Clustering Performance

Tables 3 and 4 present the results of comparative experiments conducted on ten datasets. The best result is highlighted in **bold**, while the second-best result is underlined. It is evident that the methodology introduced in this study outperforms all other compared methods across the designated metrics.

Compared to algorithms based on nonnegative matrix factorization (MultiNMF, NMF-CC, PLCMF, etc.), our algorithm based on nonnegative tensor factorization exhibits superior performance. This is because NMF-based methods typically perform matrix factorization for each view of the input multi-view data and then fuse the clustering indicator matrices only at the decision level. This ignores the potential correlations between the various views of the input data, which is crucial for multi-view data. In contrast, NTF can fully leverage the potential correlations between the various views of the anchor graph, enabling data-level fusion.

Compared to Orth-NTF, another NTF-based algorithm, our algorithm improves the clustering ACC on Scene15, Reuters, NUS, AWA, and NoisyMNIST by 0.167, 0.132, 0.152, 0.430, and 0.137, respectively. This is because our algorithm fully utilizes the probabilistic properties of anchor graphs. By applying appropriate constraints, we decompose the anchor graph into anchor labels and sample labels, achieving bi-clustering of anchors and samples. By imposing tensor rank constraints on anchor labels and sample labels, our algorithm fully considers the guiding role of anchor labels in the clustering results, which Orth-NTF does not.

5.2 Parametric Analysis

In this subsection, we examine the impact of various parameters, including the anchor rate, the value of p , and the values of λ_1 , and λ_2 on clustering performance. Figure 4 demonstrates how the anchor rate influences the seven clustering metrics. We observe that an anchor rate of approximately 0.3 yields satisfactory clustering results across all three datasets. The MSRC dataset, being smaller, requires a higher anchor rate to adequately represent its limited sample size, thereby improving clustering performance. Conversely, medium-sized datasets like HandWritten4 and Mnist4 achieve relatively excellent clustering results at anchor rates of 0.1 and 0.4, respectively. Additionally, as depicted in Figure 5, the algorithm's running time increases approximately linearly with respect to the anchor rate. Hence, the lowest feasible anchor rate is recommended, provided it meets the clustering performance criteria.

Figure 6 explores the influence of the value of p in the tensor Schatten p -norm on clustering outcomes. Notably, our model employs the same p value for both tensor Schatten p -norm regularization terms to simplify parameter tuning. Optimal clustering is achieved on the MSRC, ORL, HandWritten4, and Mnist4 datasets with p values of 0.1, 0.1, 0.9, and 0.2, respectively. This finding suggests that the tensor Schatten p -norm effectively leverages the low-rank nature of the views, enhancing the extraction of complementary information across different views and thereby improving clustering performance.

Figure 7 presents the clustering performance under various combinations of λ_1 and λ_2 . The sensitivity analysis aims to illustrate general performance trends under a

TABLE 3

Experimental performance of the compared multi-view clustering methods on MSRC, ORL, HandWritten4, Mnist4, and Scene15. Optimal results are in bold and sub-optimal results are underlined.

Datasets Methods	MSRC			ORL			HandWritten4			Mnist4			Scene15		
	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity
MultiNMF(2013)	0.757	0.662	0.790	0.710	0.854	0.803	0.803	0.678	0.803	0.818	0.631	0.818	0.369	0.380	0.396
NMF-CC(2020)	0.577	0.469	0.606	0.793	0.901	0.817	0.624	0.567	0.587	0.845	0.659	0.845	0.368	0.364	0.390
GMC(2020)	0.895	0.809	0.895	0.633	0.857	0.715	0.861	0.859	0.861	0.920	0.807	0.920	0.140	0.058	0.146
LMVSC(2021)	0.814	0.717	0.814	0.653	0.806	0.710	0.904	0.831	0.904	0.892	0.726	0.892	0.355	0.331	0.399
PLCMF(2021)	0.634	0.567	0.660	0.546	0.751	0.587	0.679	0.648	0.696	0.792	0.646	0.813	0.480	0.476	0.520
FPMVS-CAG(2021)	0.786	0.686	0.786	0.785	0.893	0.810	0.744	0.753	0.744	0.885	0.715	0.885	0.463	0.486	0.481
FMCNOF(2021)	0.713	0.648	0.714	0.346	0.535	0.371	0.541	0.484	0.540	0.686	0.513	0.695	0.263	0.249	0.258
SFMC(2022)	0.810	0.721	0.810	0.760	0.914	0.815	0.853	0.871	0.873	0.916	0.797	0.916	0.188	0.135	0.202
AWMVC(2023)	0.843	0.727	0.843	0.788	0.897	0.808	0.876	0.790	0.876	0.878	0.699	0.878	0.613	0.577	0.653
FSMSC(2023)	0.876	0.753	0.876	0.770	0.884	0.798	0.878	0.787	0.878	0.791	0.579	0.791	0.440	0.414	0.450
TBGL(2023)	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	0.995	0.995	0.995	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	0.938	0.859	0.938	0.513	0.518	0.514
Orth-NTF(2023)	0.990	0.978	0.990	0.785	0.879	0.798	0.985	0.969	0.985	0.977	0.926	0.977	0.758	0.804	0.759
LMVC-MFCG(2024)	0.852	0.738	0.852	0.653	0.815	0.688	0.856	0.832	0.856	0.886	0.717	0.886	0.505	0.566	0.522
EMVC-NTLC(2024)	0.857	0.811	0.857	0.690	0.821	0.708	0.767	0.774	0.774	0.963	0.895	0.963	0.674	0.724	0.680
AGFS-OMVC(2024)	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	0.998	0.998	0.998	0.995	0.987	0.995	0.979	0.928	0.979	0.796	0.834	0.806
ALPC(2025)	0.881	0.775	0.881	0.710	0.840	0.738	0.710	0.662	0.730	0.783	0.580	0.783	0.363	0.336	0.393
LMTC(2025)	0.881	0.808	0.881	0.830	0.916	0.851	0.914	0.840	0.914	0.883	0.707	0.883	0.587	0.583	0.640
Dual MoE MVKM(2026)	0.733	0.655	0.767	0.690	0.825	0.723	0.708	0.668	0.710	0.893	0.734	0.750	0.372	0.373	0.195
Ours	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.925	0.904	0.925

TABLE 4

Experimental performance of the compared multi-view clustering methods on Reuters, NUS, AWA, NoisyMNIST and YoutubeFace. Optimal results are in bold and sub-optimal results are underlined. 'OOM' means Out Of Memory.

Datasets Methods	Reuters			NUS			AWA			NoisyMNIST			YoutubeFace		
	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity
LMVSC(2021)	0.589	0.335	0.615	0.133	0.110	0.167	0.075	0.086	0.089	0.388	0.344	0.434	0.148	0.129	0.169
PLCMF(2021)	0.397	0.138	0.422	0.120	0.098	0.208	0.082	0.085	0.097	0.593	0.479	0.618	0.107	0.084	0.272
FPMVS-CAG(2021)	0.526	0.323	0.603	0.174	0.128	0.210	0.091	0.109	0.096	0.554	0.513	0.567	0.241	0.243	0.327
FMCNOF(2021)	0.343	0.125	0.358	0.133	0.040	0.155	0.050	0.022	0.051	0.333	0.237	0.340	0.235	0.018	0.266
SFMC(2022)	0.602	0.354	0.604	0.179	0.191	0.255	0.039	0.003	0.040	0.699	0.681	0.727	0.210	0.059	0.095
AWMVC(2023)	0.536	0.303	0.576	0.126	0.122	0.229	0.094	0.113	0.115	0.543	0.468	0.555	0.236	0.209	0.154
FSMSC(2023)	0.517	0.307	0.558	0.143	0.131	0.237	0.097	0.113	0.118	0.601	0.532	0.641	0.178	0.152	0.303
TBGL(2023)	0.795	0.660	0.795	0.274	0.202	0.281	0.050	0.023	0.053	<u>0.764</u>	<u>0.748</u>	<u>0.781</u>	OOM	OOM	OOM
Orth-NTF(2023)	0.694	0.686	0.809	0.355	0.544	0.465	0.109	0.115	0.134	0.701	0.729	0.747	0.083	0.045	0.266
LMVC-MFCG(2024)	0.579	0.313	0.586	0.183	0.135	0.219	0.097	0.118	0.104	0.546	0.458	0.552	<u>0.258</u>	0.256	<u>0.343</u>
EMVC-NTLC(2024)	0.727	0.570	0.727	0.453	0.519	0.483	0.234	0.375	0.236	0.751	0.733	0.758	0.124	0.004	0.266
AGFS-OMVC(2024)	0.646	0.593	0.771	0.300	0.467	0.390	<u>0.240</u>	<u>0.520</u>	<u>0.278</u>	0.704	0.569	0.704	OOM	OOM	OOM
ALPC(2025)	0.481	0.268	0.564	0.143	0.149	0.256	0.081	0.086	0.103	0.204	0.057	0.218	0.254	<u>0.253</u>	0.338
LMTC(2025)	0.459	0.268	0.528	0.122	0.101	0.212	0.109	0.120	0.126	0.608	0.594	0.657	0.253	0.223	0.251
Dual MoE MVKM(2026)	0.501	0.302	0.560	0.117	0.103	0.210	0.070	0.071	0.084	0.438	0.401	0.494	0.120	0.100	0.270
Ours	0.826	0.716	0.827	0.507	0.637	0.661	0.539	0.770	0.606	0.838	0.829	0.849	0.268	0.203	0.357

unified scale rather than to enumerate all optimal values. To determine the optimal values of λ_1 , and λ_2 , we first estimate their ranges based on the magnitudes of the tensor Schatten p -norm regularization terms. Subsequently, detailed fine-tuning within these ranges is conducted to refine our parameter selection.

5.3 Ablation Experiments

As shown in Table 5 and Table 6, we conducted ablation experiments on the two tensor Schatten p -norm regularization terms and the use of denoised anchor graphs. It can be observed that the tensor Schatten p -norm of the cluster labels on the samples, $\|\mathcal{H}\|_{\mathcal{S}_p}^p$, is more critical than that on the anchors, $\|\mathcal{G}\|_{\mathcal{S}_p}^p$.

The rationale behind this approach is that imposing low-rank tensor constraints on $\mathcal{H}$ effectively excavates both the complementary and spatial structural information inherent in the clustering index matrices derived from various views. Meanwhile, $\mathcal{G}$ encapsulates the clustering distribution of the anchors. The concurrent application of constraints on both $\mathcal{H}$ and $\mathcal{G}$ significantly enhances the overall clustering

efficacy. Furthermore, incorporating denoised anchor graphs improves the quality of the anchor graph and thus leads to better clustering performance.

5.4 Robustness Analysis under Shuffled Data

Recent work [52] has pointed out that some tensor-based multi-view clustering methods may potentially suffer from the inadvertent label use problem, which can lead to overly optimistic clustering performance under certain data ordering conditions. To provide a more rigorous and diagnostic evaluation, we conduct additional experiments under a shuffled-data setting, where the sample order is randomly permuted in a consistent manner across all views prior to model training.

The experimental results on shuffled datasets are reported in Table 7. It can be observed that methods not involving rotation-based tensor operations (e.g., FPMVS-CAG, Dual MoE MVKM, and LMTC) are largely insensitive to data shuffling. In contrast, methods employing rotation-based tensor modeling, including Orth-NTF and the proposed method, exhibit noticeable performance degradation after

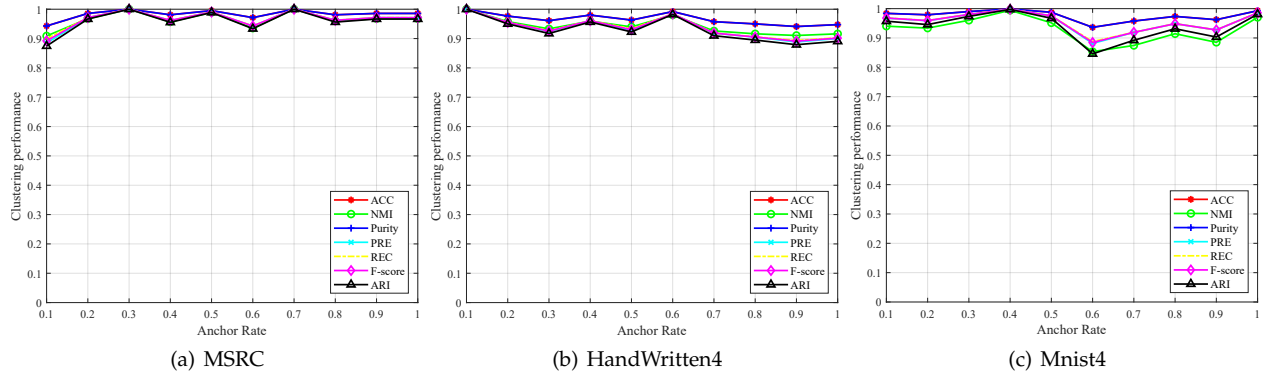


Fig. 4. Clustering performance with different anchor rate over MSRC, HandWritten4 and Mnist4.

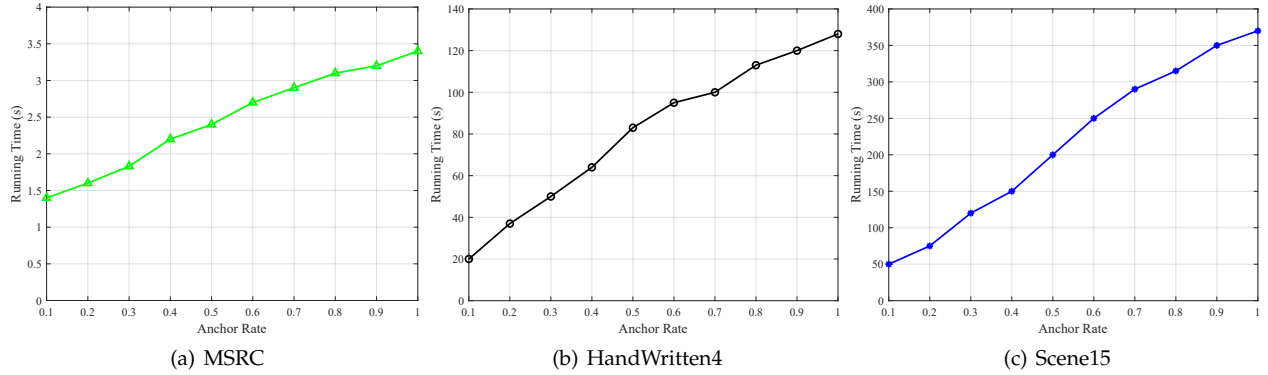


Fig. 5. Running time (sec.) with different number of anchors on MSRC, HandWritten4 and Scene15.

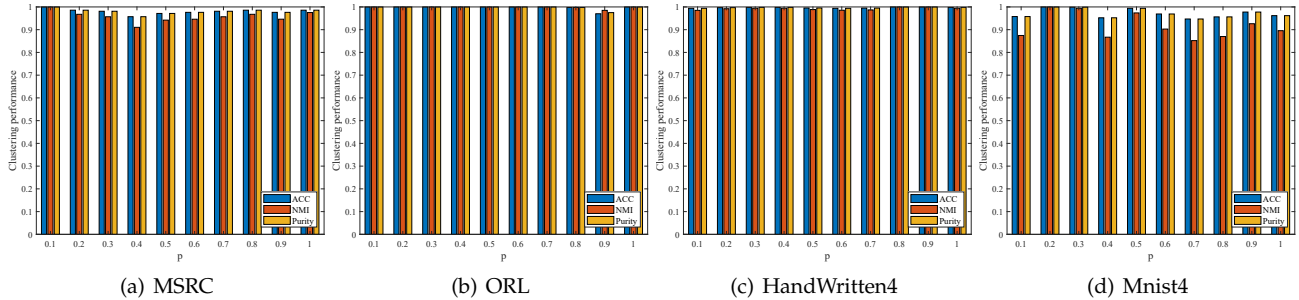


Fig. 6. Sensitivity analysis of p parameter of our method on MSRC, ORL, HandWritten4 and Mnist4.

shuffling. This observation is consistent with the empirical findings reported in [52].

This behavior is mainly related to the specific tensor construction adopted in our framework. In particular, the proposed method involves tensor rotation, followed by a `shiftdim` operation and a Fast Fourier Transform (FFT) applied along the sample-related mode. As a result, the resulting tensor representation may introduce sensitivity to sample ordering, making the shuffled-data evaluation a stricter diagnostic setting.

Overall, these results indicate that rotation-based tensor formulations may exhibit sensitivity to sample ordering. Nevertheless, the proposed framework remains competitive under shuffled-data evaluation, demonstrating its practical robustness compared with existing multi-view clustering methods.

5.5 Running time analysis

Table 8 reports the running time (in seconds) of all compared algorithms on six benchmark datasets, where the ‘-’ symbol indicates that the algorithm did not finish running within three hours (10800 seconds).

It can be seen that methods based on simple matrix factorization (e.g., NMF-CC, PLCMF) are generally the fastest, as these methods do not involve complex tensor operations.

Compared to these lightweight baseline methods, our proposed method is more computationally expensive. This is expected, as our model involves tensor construction, FFT-based tensor operations, and tensor Schatten p -norm regularization.

Compared with other subspace-based multi-view clustering methods (e.g., FPMVS-CAG and FSMSC) whose time

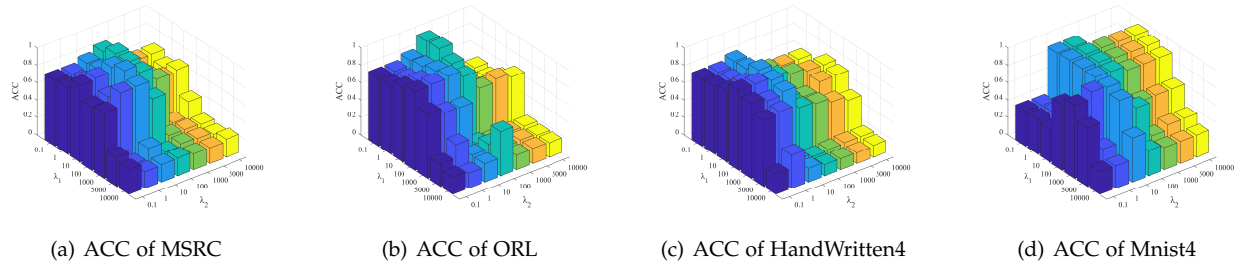


Fig. 7. The influence of λ_1 and λ_2 on clustering performance on MSRC, ORL, HandWritten4 and Mnist4.

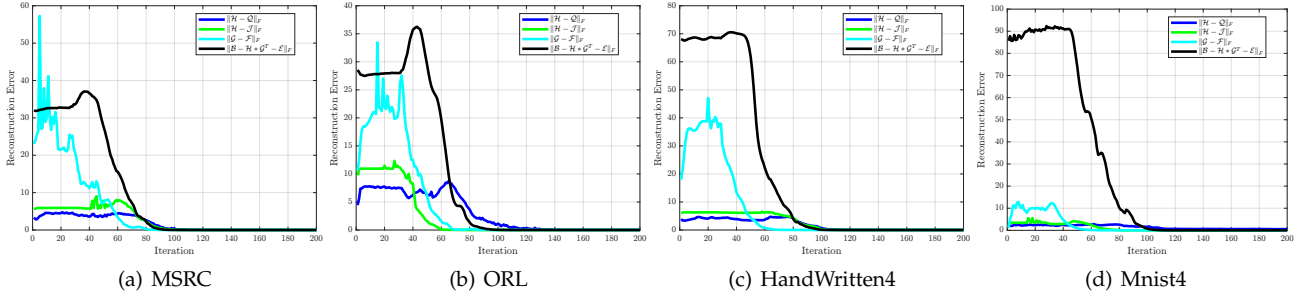


Fig. 8. Convergence of our proposed method on MSRC, ORL, HandWritten4 and Mnist4.

TABLE 5

Ablation experiments on MSRC, ORL, HandWritten4, Mnist4 and Scene15. Optimal results are in bold. 'Denoised AG' indicates the noise-reduced Anchor Graph.

Datasets			MSRC			ORL			HandWritten4			Mnist4			Scene15		
$\ \mathcal{H}\ _{\mathcal{G}}^p$	$\ \mathcal{G}\ _{\mathcal{G}}^p$	Denoised AG	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity
$\times$	$\times$	$\times$	0.789	0.713	0.789	0.822	0.894	0.848	0.811	0.769	0.811	0.914	0.786	0.914	0.499	0.490	0.541
$\times$	$\checkmark$	$\times$	0.952	0.908	0.952	0.915	0.962	0.930	0.888	0.893	0.888	0.935	0.844	0.935	0.628	0.562	0.649
$\checkmark$	$\times$	$\times$	0.971	0.943	0.971	0.880	0.939	0.893	0.905	0.874	0.905	0.956	0.865	0.956	0.837	0.860	0.858
$\checkmark$	$\checkmark$	$\times$	1.000	1.000	1.000	0.963	0.978	0.965	0.995	0.989	0.995	0.996	0.983	0.996	0.870	0.906	0.902
$\checkmark$	$\checkmark$	$\checkmark$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.925	0.904	0.925

TABLE 6

Ablation experiments on Reuters, NUS, AWA, NoisyMNIST, and YoutubeFace. Optimal results are in bold. 'Denoised AG' indicates the noise-reduced Anchor Graph.

Datasets			Reuters			NUS			AWA			NoisyMNIST			YoutubeFace		
$\ \mathcal{H}\ _{\mathcal{G}}^p$	$\ \mathcal{G}\ _{\mathcal{G}}^p$	Denoised AG	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity
$\times$	$\times$	$\times$	0.421	0.158	0.489	0.113	0.092	0.120	0.066	0.067	0.082	0.530	0.415	0.531	0.130	0.093	0.274
$\times$	$\checkmark$	$\times$	0.436	0.128	0.482	0.131	0.077	0.192	0.066	0.069	0.081	0.621	0.509	0.645	0.142	0.108	0.280
$\checkmark$	$\times$	$\times$	0.715	0.666	0.793	0.447	0.532	0.572	0.455	0.669	0.495	0.717	0.709	0.730	0.180	0.135	0.301
$\checkmark$	$\checkmark$	$\times$	0.796	0.731	0.796	0.442	0.614	0.580	0.510	0.704	0.560	0.819	0.838	0.851	0.224	0.182	0.319
$\checkmark$	$\checkmark$	$\checkmark$	0.826	0.716	0.827	0.507	0.637	0.661	0.539	0.770	0.606	0.838	0.829	0.849	0.268	0.203	0.357

complexity scales linearly with the number of samples, the running time of our algorithm is comparable to these methods across all benchmark datasets.

Despite the additional computational overhead, our proposed method remains practically feasible on all benchmark datasets without excessive runtime inflation. This allows the proposed method to achieve a good balance between clustering performance and computational efficiency.

5.6 Experiments of Convergence

To optimize the objective function (10), we iteratively introduce three auxiliary variables: $\mathcal{Q}$, $\mathcal{G}$, and $\mathcal{F}$. The model's convergence is assessed by analyzing the reconstruction error:

$$\|\mathcal{H} - \mathcal{Q}\|_F, \|\mathcal{H} - \mathcal{J}\|_F, \|\mathcal{G} - \mathcal{F}\|_F, \text{ and } \|\mathcal{B} - \mathcal{H} * \mathcal{G}^T - \mathcal{E}\|_F.$$

Fig. 8 illustrates the variations of reconstruction error. Convergence is typically achieved around the 120th iteration, where the differences stabilize, indicating a steady state.

5.7 Visualization of Experimental Results

Fig. 11 presents the clustering results through t-SNE plots of our algorithm, revealing distinct and well-separated clusters. Note that the t-SNE visualization is provided for qualitative analysis only. Due to the nonlinear projection from high-dimensional space to two dimensions, apparent overlaps in the visualization do not necessarily indicate incorrect clustering assignments in the original latent space. The

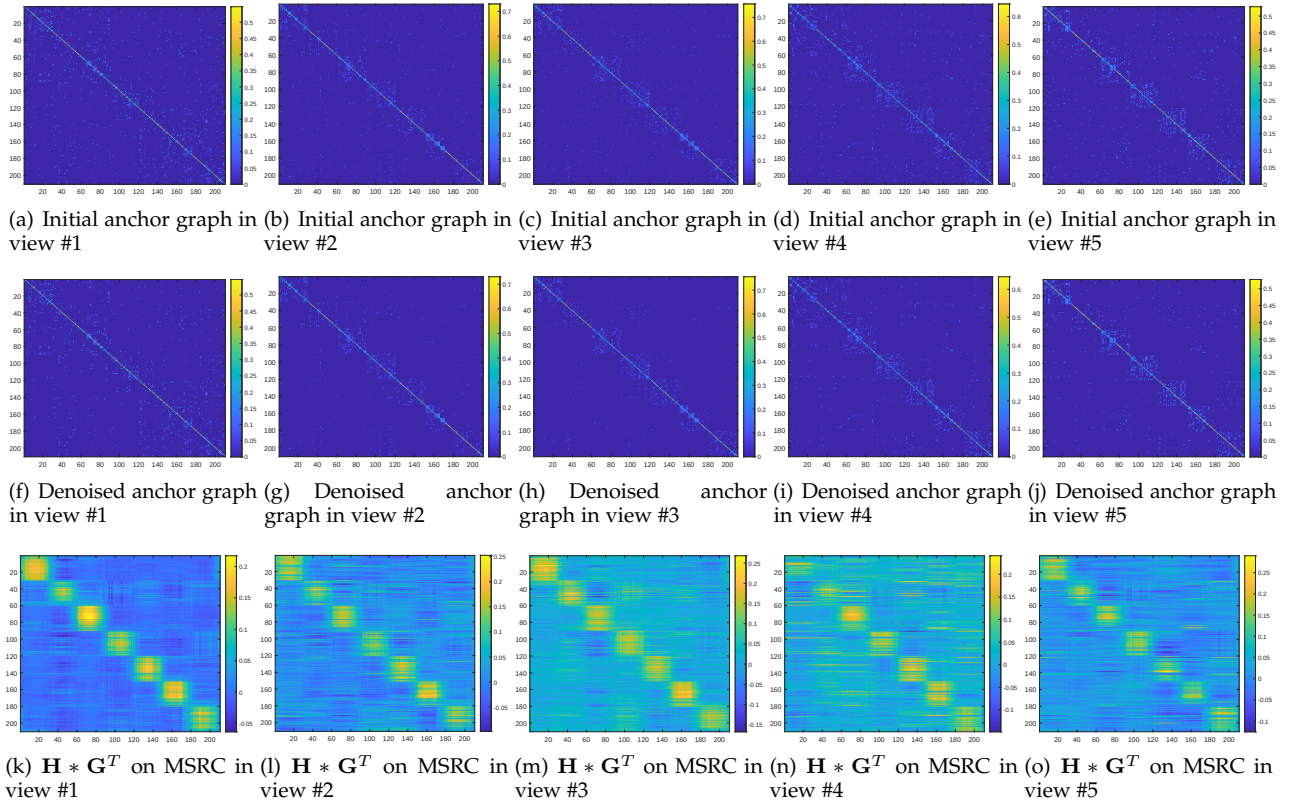


Fig. 9. Visualization of initial anchor graphs and learned denoised anchor graphs on the MSRC dataset

TABLE 7

The comparison algorithm's clustering result (ACC) on shuffled datasets are shown, where $\uparrow$ and $\downarrow$ represent performance improvements and decreases, respectively.

Datasets	FPMVS-CAG			Orth-NTF			Dual MoE MVKM			LMTC			Ours		
	Sort	Shuffle	Gap	Sort	Shuffle	Gap	Sort	Shuffle	Gap	Sort	Shuffle	Gap	Sort	Shuffle	Gap
MSRC	0.786	0.733	0.053 $\downarrow$	0.990	0.609	0.381 $\downarrow$	0.733	0.681	0.052 $\downarrow$	0.881	0.881	0.000	1.000	0.824	0.176 $\downarrow$
ORL	0.785	0.657	0.128 $\downarrow$	0.785	0.638	0.147 $\downarrow$	0.690	0.722	0.032 $\uparrow$	0.830	0.813	0.017 $\downarrow$	1.000	0.875	0.125 $\downarrow$
HandWritten4	0.744	0.648	0.096 $\downarrow$	0.985	0.461	0.524 $\downarrow$	0.708	0.714	0.006 $\uparrow$	0.914	0.912	0.002 $\downarrow$	1.000	0.839	0.161 $\downarrow$
Mnist4	0.885	0.885	0.000	0.977	0.654	0.323 $\downarrow$	0.893	0.892	0.001 $\downarrow$	0.883	0.883	0.000	1.000	0.918	0.082 $\downarrow$

TABLE 8

Running time (s) of compared algorithms on six benchmark datasets. '-' means that the algorithm has been running for more than three hours.

	HandWritten4	Mnist4	NUS	AWA	NoisyMNIST
MultiNMF	9.37	5.80	1172.88	-	-
NMF-CC	0.67	0.57	61.82	303.46	365.94
LMVSC	2.54	3.71	45.37	161.45	118.34
PLCMF	0.58	0.63	13.43	75.20	175.95
FPMVS-CAG	13.05	23.26	652.84	1844.47	936.77
AWMVS	2.04	2.14	131.73	1308.41	469.54
FSMSC	14.45	28.27	799.41	1286.99	1169.11
LMVC-MFCG	30.28	10.26	487.15	339.56	763.19
ALPC	3.26	1.66	321.92	939.95	931.24
Dual MoE MVKM	1.30	0.30	24.24	955.72	115.56
LMTC	0.30	0.30	240.29	503.77	121.96
Ours	25.63	33.02	520.48	1061.36	654.95

learned labels are visualized in Figure 10. Additionally, the input anchor graph and the learned denoised anchor graph for the MSRC dataset are depicted in Fig. 9. It can be observed that the original anchor graph contains noticeable noisy connections across different clusters, while the denoised anchor graph exhibits a clearer block-diagonal structure.

These visualizations indicate that our method effectively captures the clustering structure of the data, leveraging the complementarities among multiple views.

6 CONCLUSION

In this paper, we perform non-negative tensor factorization (NTF) on tensorized anchor graph which are constructed from multiple view-specific anchor graphs, and then build orthogonal NTF clustering model which achieves both data-level and decision-level fusion. This simultaneous processing of all view data ensures the comprehensive utilization of inter-view information. Furthermore, we demonstrate that the tensors resulting from this decomposition represent the clustering of both samples and anchors, thereby enhancing

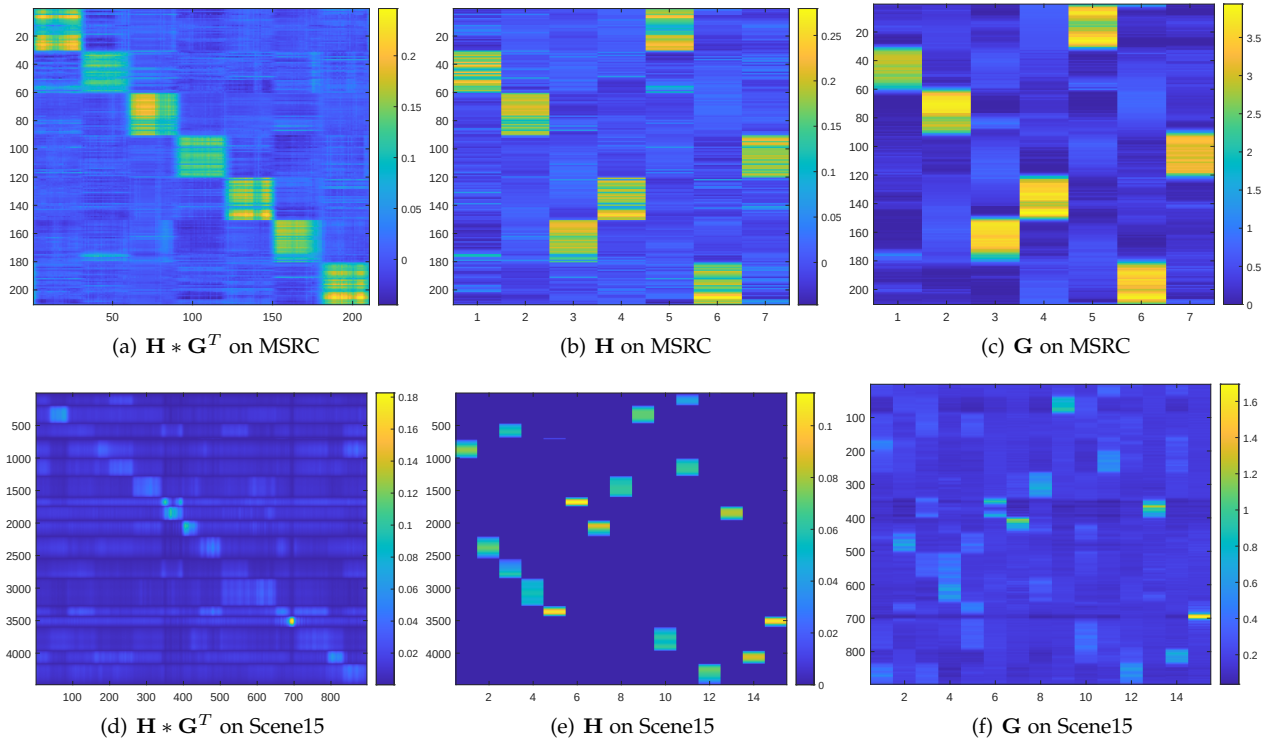


Fig. 10. The first three columns (from left to right) correspond to the visualization of the fused $H * G^T$, sample clustering label H , and anchor clustering label G , where each row corresponds to one of the two datasets (MSRC and Scene15).

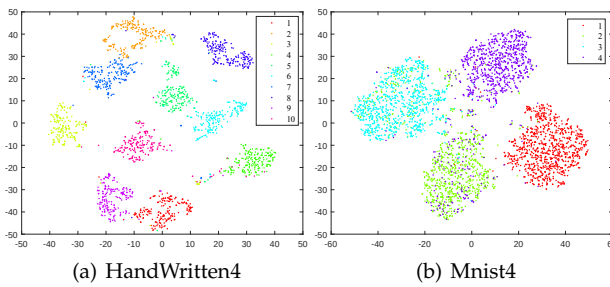


Fig. 11. Visualization of latent representation with t-SNE on HandWritten4 and Mnist4.

the interpretability of the decomposition. Extensive experimental evaluations validate the effectiveness of the proposed methodology.

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